STA ANALYSIS — SKY130 HD

Static Timing Analysis of All Provided Timing Corners

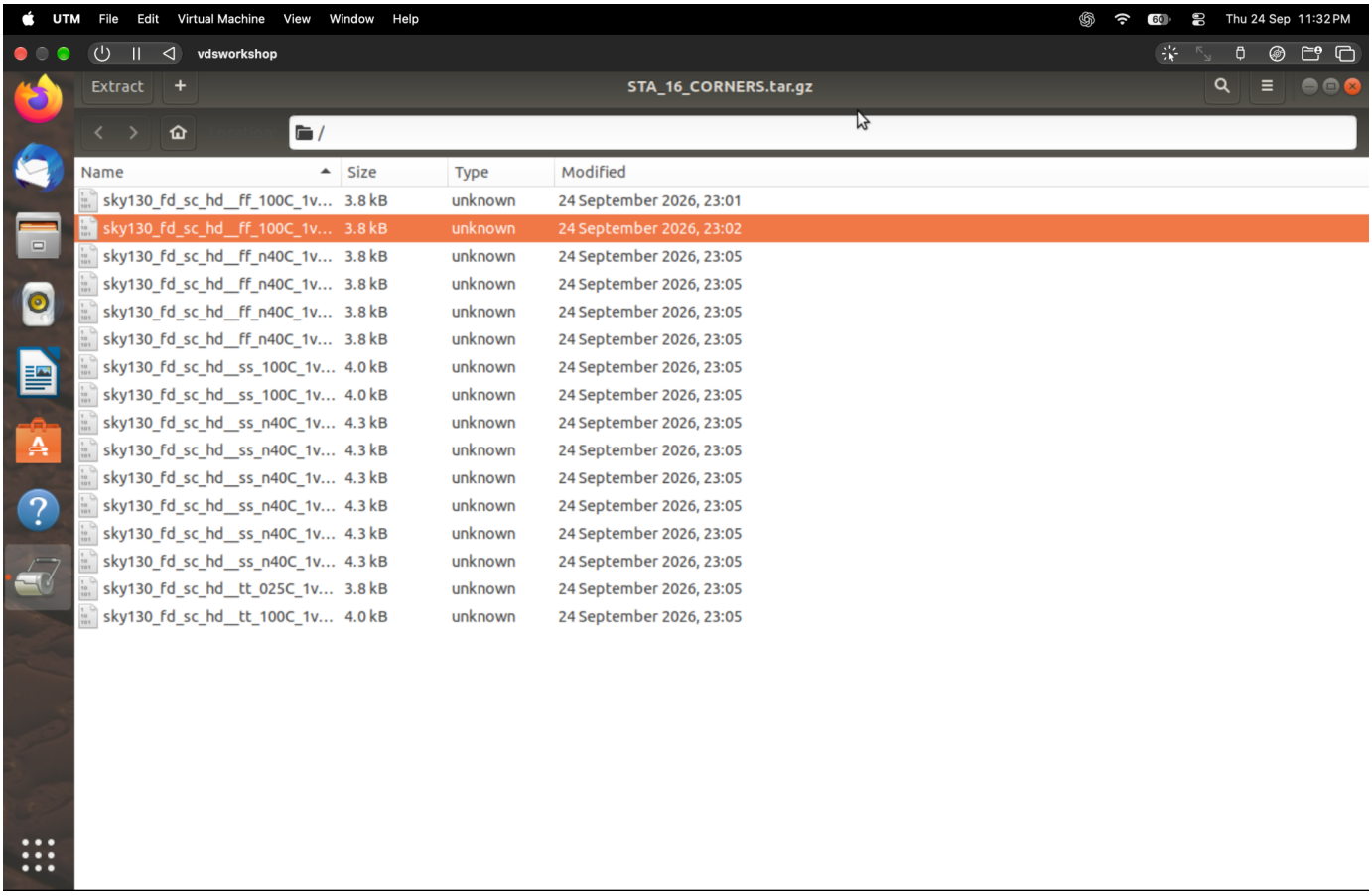
Roll No.: 24EG104E28

Execution

The STA was executed independently on my Ubuntu/QTM environment using the SKY130 HD timing libraries.
The following screenshot is the actual output from my run and shows the generated reports for all 16 provided corners.

Important commands used

```
read_liberty <corner_library.lib>
read_verilog vsdbabysoc.synth.v
link_design vsdbabysoc
read_sdc vsdbabysoc_synthesis.sdc
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}
```

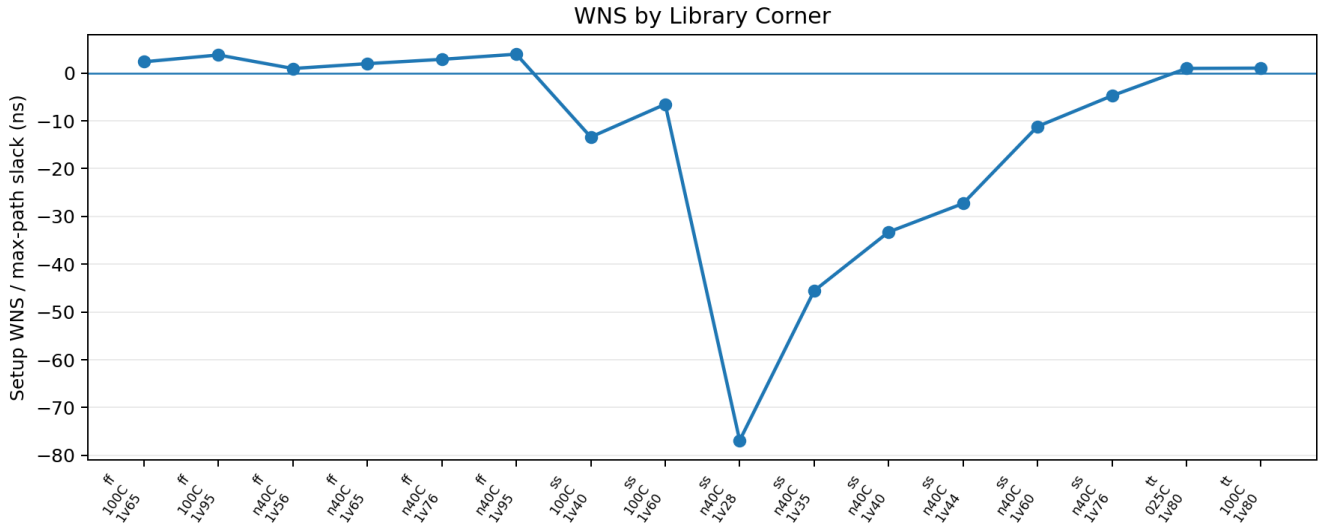


What the screenshot represents

- The archive contains the STA report generated for each of the 16 SKY130 HD timing corners.
- Each report corresponds to a different PVT condition: process (FF/SS/TT), temperature and supply voltage.
- The corner names visible in the screenshot identify the individual libraries used for the STA runs.
- These reports were generated from the same synthesized VSDBabySoC netlist and SDC constraints.
- Detailed path analysis was obtained using report_checks with min/max path delay and the requested nets, capacitance, slew, input-pin and fanout fields.
- The results in the following pages are taken from these actual generated reports.

16-Corner Result Summary

#	Library corner	Process / Temp / V	Min slack (ns)	Setup slack (ns)	Status
1	ff_100C_1v65	FF / +100 °C / 1.65 V	0.25	2.35	MET
2	ff_100C_1v95	FF / +100 °C / 1.95 V	0.20	3.77	MET
3	ff_n40C_1v56	FF / -40 °C / 1.56 V	0.29	0.94	MET
4	ff_n40C_1v65	FF / -40 °C / 1.65 V	0.26	1.96	MET
5	ff_n40C_1v76	FF / -40 °C / 1.76 V	0.22	2.86	MET
6	ff_n40C_1v95	FF / -40 °C / 1.95 V	0.19	3.94	MET
7	ss_100C_1v40	SS / +100 °C / 1.40 V	0.91	-13.40	VIOLATED
8	ss_100C_1v60	SS / +100 °C / 1.60 V	0.64	-6.54	VIOLATED
9	ss_n40C_1v28	SS / -40 °C / 1.28 V	1.83	-76.89	VIOLATED
10	ss_n40C_1v35	SS / -40 °C / 1.35 V	1.35	-45.57	VIOLATED
11	ss_n40C_1v40	SS / -40 °C / 1.40 V	1.12	-33.27	VIOLATED
12	ss_n40C_1v44	SS / -40 °C / 1.44 V	0.99	-27.31	VIOLATED
13	ss_n40C_1v60	SS / -40 °C / 1.60 V	0.66	-11.18	VIOLATED
14	ss_n40C_1v76	SS / -40 °C / 1.76 V	0.50	-4.76	VIOLATED
15	tt_025C_1v80	TT / +25 °C / 1.80 V	0.31	0.96	MET
16	tt_100C_1v80	TT / +100 °C / 1.80 V	0.31	1.02	MET



Negative setup slack indicates a setup timing violation for the reported worst max path.

ff_100C_1v65

Meaning: Fast NMOS / Fast PMOS; temperature +100 °C; supply 1.65 V. This corner represents the specified PVT condition used for timing characterization.

Result: min/hold slack 0.25 ns; setup/max-path slack 2.35 ns — **MET**.

```
OpenSTA 2.4.0
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

MIN / HOLD PATH
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Type: min
Reported min slack:  0.25 ns (MET)

MAX / SETUP PATH
Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10042_ (rising edge-triggered flip-flop clocked by clk)
Path Type: max
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
Reported setup/max slack: 2.35 ns (MET)
```

ff_100C_1v95

Meaning: Fast NMOS / Fast PMOS; temperature +100 °C; supply 1.95 V. This corner represents the specified PVT condition used for timing characterization.

Result: min/hold slack 0.20 ns; setup/max-path slack 3.77 ns — **MET**.

```
OpenSTA 2.4.0
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

MIN / HOLD PATH
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Type: min
Reported min slack:  0.20 ns (MET)

MAX / SETUP PATH
Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10042_ (rising edge-triggered flip-flop clocked by clk)
Path Type: max
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
Reported setup/max slack: 3.77 ns (MET)
```

ff_n40C_1v56

Meaning: Fast NMOS / Fast PMOS; temperature -40 °C; supply 1.56 V. This corner represents the specified PVT condition used for timing characterization.

Result: min/hold slack 0.29 ns; setup/max-path slack 0.94 ns — **MET**.

```
OpenSTA 2.4.0
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

MIN / HOLD PATH
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Type: min
Reported min slack:  0.29 ns (MET)

MAX / SETUP PATH
Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10042_ (rising edge-triggered flip-flop clocked by clk)
Path Type: max
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
Reported setup/max slack: 0.94 ns (MET)
```

ff_n40C_1v65

Meaning: Fast NMOS / Fast PMOS; temperature -40 °C; supply 1.65 V. This corner represents the specified PVT condition used for timing characterization.

Result: min/hold slack 0.26 ns; setup/max-path slack 1.96 ns — **MET**.

```
OpenSTA 2.4.0
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

MIN / HOLD PATH
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Type: min
Reported min slack:  0.26 ns (MET)

MAX / SETUP PATH
Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10042_ (rising edge-triggered flip-flop clocked by clk)
Path Type: max
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
Reported setup/max slack: 1.96 ns (MET)
```

ff_n40C_1v76

Meaning: Fast NMOS / Fast PMOS; temperature -40 °C; supply 1.76 V. This corner represents the specified PVT condition used for timing characterization.

Result: min/hold slack 0.22 ns; setup/max-path slack 2.86 ns — **MET**.

```
OpenSTA 2.4.0
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

MIN / HOLD PATH
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Type: min
Reported min slack:  0.22 ns (MET)

MAX / SETUP PATH
Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10042_ (rising edge-triggered flip-flop clocked by clk)
Path Type: max
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
Reported setup/max slack: 2.86 ns (MET)
```

ff_n40C_1v95

Meaning: Fast NMOS / Fast PMOS; temperature -40 °C; supply 1.95 V. This corner represents the specified PVT condition used for timing characterization.

Result: min/hold slack 0.19 ns; setup/max-path slack 3.94 ns — **MET**.

```
OpenSTA 2.4.0
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

MIN / HOLD PATH
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Type: min
Reported min slack:  0.19 ns (MET)

MAX / SETUP PATH
Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10042_ (rising edge-triggered flip-flop clocked by clk)
Path Type: max
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
Reported setup/max slack: 3.94 ns (MET)
```

ss_100C_1v40

Meaning: Slow NMOS / Slow PMOS; temperature +100 °C; supply 1.40 V. This corner represents the specified PVT condition used for timing characterization.

Result: min/hold slack 0.91 ns; setup/max-path slack -13.40 ns — **VIOLATED**.

```
OpenSTA 2.4.0
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

MIN / HOLD PATH
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Type: min
Reported min slack:  0.91 ns (MET)

MAX / SETUP PATH
Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10233_ (rising edge-triggered flip-flop clocked by clk)
Path Type: max
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
Reported setup/max slack: -13.40 ns (VIOLATED)
```

ss_100C_1v60

Meaning: Slow NMOS / Slow PMOS; temperature +100 °C; supply 1.60 V. This corner represents the specified PVT condition used for timing characterization.

Result: min/hold slack 0.64 ns; setup/max-path slack -6.54 ns — **VIOLATED**.

```
OpenSTA 2.4.0
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

MIN / HOLD PATH
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Type: min
Reported min slack:  0.64 ns (MET)

MAX / SETUP PATH
Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10233_ (rising edge-triggered flip-flop clocked by clk)
Path Type: max
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
Reported setup/max slack: -6.54 ns (VIOLATED)
```

ss_n40C_1v28

Meaning: Slow NMOS / Slow PMOS; temperature -40 °C; supply 1.28 V. This corner represents the specified PVT condition used for timing characterization.

Result: min/hold slack 1.83 ns; setup/max-path slack -76.89 ns — **VIOLATED**.

```
OpenSTA 2.4.0
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

MIN / HOLD PATH
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Type: min
Reported min slack:  1.83 ns (MET)

MAX / SETUP PATH
Startpoint: _10277_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10321_ (rising edge-triggered flip-flop clocked by clk)
Path Type: max
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ _10277_/CLK (sky130_fd_sc_hd__dfxtp_1)
Reported setup/max slack: -76.89 ns (VIOLATED)
```

ss_n40C_1v35

Meaning: Slow NMOS / Slow PMOS; temperature -40 °C; supply 1.35 V. This corner represents the specified PVT condition used for timing characterization.

Result: min/hold slack 1.35 ns; setup/max-path slack -45.57 ns — **VIOLATED**.

```
OpenSTA 2.4.0
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

MIN / HOLD PATH
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Type: min
Reported min slack:  1.35 ns (MET)

MAX / SETUP PATH
Startpoint: _10277_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10321_ (rising edge-triggered flip-flop clocked by clk)
Path Type: max
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ _10277_/CLK (sky130_fd_sc_hd__dfxtp_1)
Reported setup/max slack: -45.57 ns (VIOLATED)
```

ss_n40C_1v40

Meaning: Slow NMOS / Slow PMOS; temperature -40 °C; supply 1.40 V. This corner represents the specified PVT condition used for timing characterization.

Result: min/hold slack 1.12 ns; setup/max-path slack -33.27 ns — **VIOLATED**.

```
OpenSTA 2.4.0
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

MIN / HOLD PATH
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Type: min
Reported min slack:  1.12 ns (MET)

MAX / SETUP PATH
Startpoint: _10277_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10321_ (rising edge-triggered flip-flop clocked by clk)
Path Type: max
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ _10277_/CLK (sky130_fd_sc_hd__dfxtp_1)
Reported setup/max slack: -33.27 ns (VIOLATED)
```

ss_n40C_1v44

Meaning: Slow NMOS / Slow PMOS; temperature -40 °C; supply 1.44 V. This corner represents the specified PVT condition used for timing characterization.

Result: min/hold slack 0.99 ns; setup/max-path slack -27.31 ns — **VIOLATED**.

```
OpenSTA 2.4.0
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

MIN / HOLD PATH
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Type: min
Reported min slack:  0.99 ns (MET)

MAX / SETUP PATH
Startpoint: _10277_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10321_ (rising edge-triggered flip-flop clocked by clk)
Path Type: max
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ _10277_/CLK (sky130_fd_sc_hd__dfxtp_1)
Reported setup/max slack: -27.31 ns (VIOLATED)
```

ss_n40C_1v60

Meaning: Slow NMOS / Slow PMOS; temperature -40 °C; supply 1.60 V. This corner represents the specified PVT condition used for timing characterization.

Result: min/hold slack 0.66 ns; setup/max-path slack -11.18 ns — **VIOLATED**.

```
OpenSTA 2.4.0
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

MIN / HOLD PATH
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Type: min
Reported min slack:  0.66 ns (MET)

MAX / SETUP PATH
Startpoint: _10277_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10321_ (rising edge-triggered flip-flop clocked by clk)
Path Type: max
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ _10277_/CLK (sky130_fd_sc_hd__dfxtp_1)
Reported setup/max slack: -11.18 ns (VIOLATED)
```

ss_n40C_1v76

Meaning: Slow NMOS / Slow PMOS; temperature -40 °C; supply 1.76 V. This corner represents the specified PVT condition used for timing characterization.

Result: min/hold slack 0.50 ns; setup/max-path slack -4.76 ns — **VIOLATED**.

```
OpenSTA 2.4.0
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

MIN / HOLD PATH
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Type: min
Reported min slack:  0.50 ns (MET)

MAX / SETUP PATH
Startpoint: _10277_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10321_ (rising edge-triggered flip-flop clocked by clk)
Path Type: max
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ _10277_/CLK (sky130_fd_sc_hd__dfxtp_1)
Reported setup/max slack: -4.76 ns (VIOLATED)
```

tt_025C_1v80

Meaning: Typical NMOS / Typical PMOS; temperature +25 °C; supply 1.80 V. This corner represents the specified PVT condition used for timing characterization.

Result: min/hold slack 0.31 ns; setup/max-path slack 0.96 ns — **MET**.

```
OpenSTA 2.4.0
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

MIN / HOLD PATH
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Type: min
Reported min slack:  0.31 ns (MET)

MAX / SETUP PATH
Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10042_ (rising edge-triggered flip-flop clocked by clk)
Path Type: max
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
Reported setup/max slack: 0.96 ns (MET)
```

tt_100C_1v80

Meaning: Typical NMOS / Typical PMOS; temperature +100 °C; supply 1.80 V. This corner represents the specified PVT condition used for timing characterization.

Result: min/hold slack 0.31 ns; setup/max-path slack 1.02 ns — **MET**.

```
OpenSTA 2.4.0
report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

MIN / HOLD PATH
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Type: min
Reported min slack:  0.31 ns (MET)

MAX / SETUP PATH
Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint:   _10233_ (rising edge-triggered flip-flop clocked by clk)
Path Type: max
                0.00    0.00    0.00   clock clk (rise edge)
                        0.00    0.00   clock network delay (ideal)
                0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
Reported setup/max slack: 1.02 ns (MET)
```

Conclusion

The 16 supplied SKY130 HD timing corners were analyzed with the required detailed `report_checks` command. The reported setup slack is positive for the FF and TT corners, while the SS corners at the listed conditions include setup violations. The complete corner-wise values are summarized above.